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14.1. EARLY "TRANSURANIUM" ELEMENTS

After the discovery of the neutron by Chadwick in 1932, the group led by Fermi in Rome began bombarding different elements with neutrons to study the radioactivity induced through (n,γ) reactions. The decay characteristics of the radioactivity in neutron-irradiated uranium was interpreted to indicate that some of the products were probably transuranium elements. Similar conclusions were reached by other research groups; for example in 1937 Meitner, Hahn and Strassman in Berlin proposed the following reaction/decay series (omitting mass numbers):

$$_{92}\mathrm{U} + \mathrm{n} \rightarrow {}_{92}\mathrm{U} \xrightarrow[10\ \mathrm{s}]{\beta} {}_{93}\mathrm{EkaRe} \xrightarrow[2.2\ \mathrm{m}]{\beta} {}_{94}\mathrm{EkaOs} \xrightarrow[59\ \mathrm{m}]{\beta} \text{etc} \qquad (14.1)$$

However, many anomalies existed in the chemical properties of these "transuranium elements" as pointed out by Ida Noddack, Irene Curie, Savitch and others.

In 1939 Hahn and Strassman conducted a series of extremely careful chemical investigations which showed that these "transuranium elements" were in fact isotopes of elements in the middle of the Periodic Table, such as Sr, La, Ba etc. The suggestion that their observations were due to a fission of uranium was first made in a scientific paper by Ida Noddack, but ignored. About two years later Meitner and Frisch postulated that the uranium atoms were caused to split by thermal neutrons into two approximately equal fragments.

Further investigation showed that it was the isotope $^{235}\mathrm{U}$ which had undergone fission, and that large amounts of energy and about 2.5 neutrons were released in the process. It occurred to a number of scientists that, if at least one of these neutrons could be captured by other uranium atoms to cause further fission, a chain reaction should be the consequence. This led to the development of nuclear energy and nuclear weapons, as described in Chapter 19.

To the present time 21 transuranium elements have been synthesized; the isotopes known are summarized in Figure 14.1; except elements above 114, which are still uncertain.

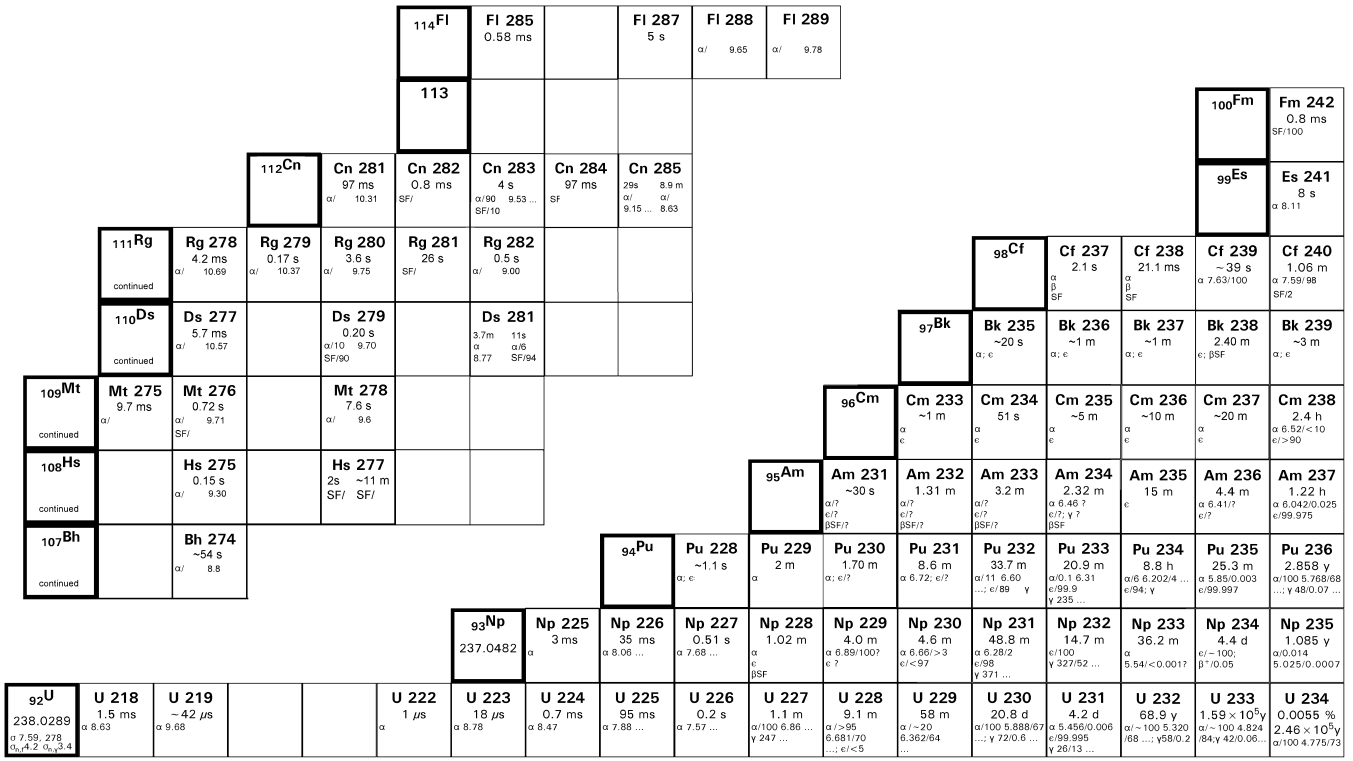


Figure 14.1 Isotope chart of uranium, transuranium, and transactinide elements.

Figure 14.1 (*continued*).



Figure 14.1 (*continued*).

14.2. THE ACTINIDE ELEMENTS

14.2.1. Actinium, element 89

Actinium was discovered in 1899 by Debierne in residues from the work of the Curies. It was named after the Greek word for beam or ray, aktis. Actinium occurs in nature as a member of the long natural decay series, except the one beginning with ^{238}U, see Fig. 1.2. Being the first of them, it has given name to the actinide series of elements. Chemically actinium behaves very similar to the lanthanides and especially as lanthanum.

14.2.1.1. Isotopes

Actinium only occurs in nature as the isotope ^{227}Ac ($t_{½}$ 21.77 y), which is a member of the long decay series starting with ^{235}U. This is also the most stable actinium isotope.

14.2.1.2. Occurrence and production

Due to its very low concentration, actinium is almost never produced from the ^{235}U decay series. However, it has been produced in macro amounts by neuron irradiation of ^{226}Ra:

$$^{226}_{88}\mathrm{Ra} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{227}_{88}\mathrm{Ra} \rightarrow {}^{227}_{89}\mathrm{Ac} + \beta^{-} \qquad (14.2)$$

14.2.1.3. Uses

Actinium has been used in radioactive n-sources made by compressing an intimate mixture of ^{227}Ac and beryllium. The neutrons are produced when the α-particles from actinium hit atoms of ^{9}Be. It was earlier suggested that actinium compounds might be used in radiation therapy of cancers, however, better alternatives now exist.

14.2.2. Thorium, element 90

Thorium was discovered by Berzelius in 1828 from a sample of what is now called thorite ($ThSiO_4$) and named by him after the nordic god of thunder, Thor. As a metal it is silvery white but with exposure to air it tarnishes slowly becoming gray and finally black.

14.2.2.1. Isotopes

Thorium occurs in nature as the isotopes ^{227,228,229,230,231,231,232,234}Th, most of them short lived. Natural thorium consists almost to 100% by weight of the very long-lived isotope ^{232}Th ($t_{½}$ 1.405×10^{10} y), which is the parent nuclide of the thorium decay series. The specific radioactivity for thorium is lower than that of uranium, and it is often treated as a non-radioactive element. For radioactive tracer studies, the nuclide ^{234}Th ($t_{½}$ 24.1 d) is usually separated from natural uranium.